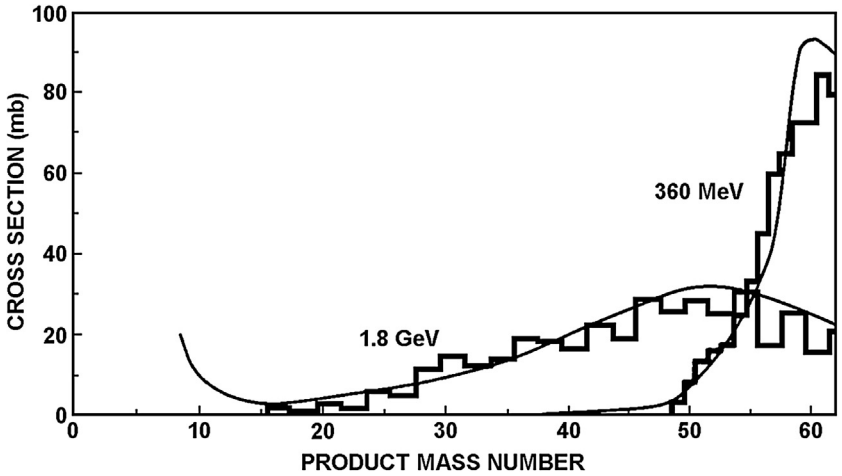


Figure 11.8 Comparison of mass yield curves for proton interaction with copper as predicted by Monte Carlo calculations (histogram) with experimental results (curve). (From G. Friedlander.)

In a *Monte Carlo calculation*, the history of each incident nucleon is studied in all its collisions in the nucleus. Each collision with a target nucleon is characterized by probability distributions for occurrence, for energy, for angular distribution, and for pion formation. The outcome of each process is obtained from a series of properly distributed random numbers and scores are accumulated for each process. The calculations are repeated for different impact points. It is the use of random numbers that is the basis of the Monte Carlo technique. Modern computers allow so many random number calculations that a useful pattern of events emerges from the cumulative scores.

Figure 11.8 shows the quite satisfactory agreement for two bombardment energies between experimental yields of A in proton bombardment of copper (solid lines) and a Monte Carlo calculation (histograms).

11.7. NUCLEAR FISSION

Nuclear fission can be either spontaneous or induced by particle (e.g. neutron) bombardment. It is fairly unique among nuclear reactions since the nucleus divides into a few heavy parts with the release of a large amount of energy, about 200 MeV per fission. It is probably no overstatement to say that fission is the most important nuclear process among man made ones, both for its potential to destroy civilization through the use of weapons and its potential through reactors to supply abundant power for all people.

An aspect of fission which has been studied extensively is the distribution of mass, charge and kinetic energy among the fragments formed in fission. No matter how nuclei are made to undergo fission, fragments of various masses are formed which result in production of chemical elements as light as zinc (atomic number 30) and as heavy as

gadolinium (atomic number 64), with half-lives from fractions of seconds to millions of years. Approximately 400 different nuclides have been identified as products in the fission of ^{235}U by neutrons. Study of these fission products has required extensive radiochemical work and continues as a very active field of research – for example, in the measurement of very short-lived products. Although fission is an extremely complicated process, and still challenges theorists, satisfactory models for most of the fission phenomena have been developed.

11.7.1. Mass and charge distribution

If fission was symmetric, i.e. if two fragments of equal mass (and charge) were formed, the thermal neutron fission of $^{235}_{92}$U would lead to the production of two $^{118}_{46}$Pd nuclides,

$$^{235}_{92}\mathrm{U} + \mathrm{n} \rightarrow 2\,^{118}_{46}\mathrm{Pd} \tag{11.15}$$

However, a plot showing the amount of different masses formed is a curve with two maxima – one near mass number 97 (close to the n-shell $N = 50$) and a second near mass number 137 (close to the n-shell $N = 82$) (Fig. 11.9). These two masses are formed together in the most probable split which is asymmetric ($A_1 \neq A_2$). As Figure 11.9 shows, symmetric fission ($A_1 = A_2$) is rare in fission of ^{235}U by thermal neutrons – the yield for $A = 115$ is only 0.01% compared to 6% for $A_1 = 97$ (or $A_2 = 137$). Since two fission products are formed in each fission event, mass yield curves like that in Figure 11.9 must total 200% on a number basis.

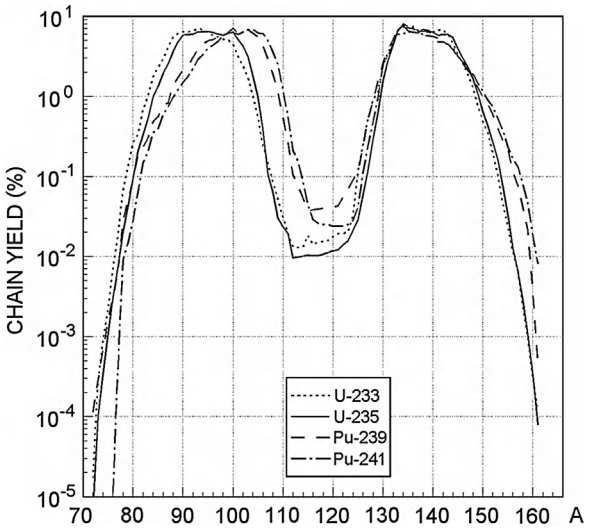


Figure 11.9 Chain yield curves for fission of ^{233}U, ^{235}U, ^{239}Pu and ^{241}Pu with thermal neutrons.

From the mass yield curve in Figure 11.9 we learn that complementary fission products (i.e. the two products A_1 and A_2 with identical yield values symmetrically located around the minimum) add up to about 234, not 236. Direct neutron measurements reveal that on the average 2.5 neutrons are emitted in fission of ^{235}U by thermal neutrons. This number increases as the Z of the target and as the bombarding energy increases.

Because the N/Z ratio for $^{236}_{92}$U(the fissioning nucleus) is 1.57, while the ratio necessary for stability is $1.2 - 1.4$ in the elements produced in fission, fission fragments always have a too large N/Z ratio. This is partially compensated by the emission of several neutrons in the act of fission, *prompt neutrons*. However, the number of neutrons emitted is not sufficient to lower the N/Z ratios to stable values. To achieve further lowering, the fission fragments, after neutron emission, undergo a series of radioactive decay steps in which β^--particles are emitted. Since the β-decay occurs with no change in A, successive β-decay steps follow the isobar parabola of the stability valley (see Fig. 4.5; cf. also the nuclide chart, Appendix C). For $A = 137$, $^{137}_{52}$Te is the first nuclide measured. The chain sequence is

$$^{137}\mathrm{Te} \xrightarrow{3.5\ \mathrm{s}} {}^{137}\mathrm{I} \xrightarrow{24.5\ \mathrm{s}} {}^{137}\mathrm{Xe} \xrightarrow{3.82\ \mathrm{min}} {}^{137}\mathrm{Cs} \xrightarrow{30.0\ \mathrm{y}} {}^{137\mathrm{m}}\mathrm{Ba} \xrightarrow{2.6\ \mathrm{m}} {}^{137}\mathrm{Ba(stable)} \tag{11.16}$$

Thermal fission of ^{235}U leads to a yield of 6.183% for the $A = 137$ chain. In a small number of cases the decay chains passes nuclides which emit a neutron after β^--decay, so called *delayed neutrons* (e.g. 0.016% of all neutrons from thermal fission of ^{235}U are delayed neutrons). Neutrons are emitted in 2% of the ^{137}Te β-decays and in 6.4% of the ^{137}I β-decays. The existence of delayed neutrons is important for nuclear reactor control, see Ch. 19.

In addition to measuring the variation of mass yield, the variation of fission yield in isobaric mass chains as a function of the proton number has been studied. In Figure 11.10, "individual" yield data are presented for the $A = 93$ chain. In general, the charge distribution yields follow a Gaussian curve with the maximum displaced several

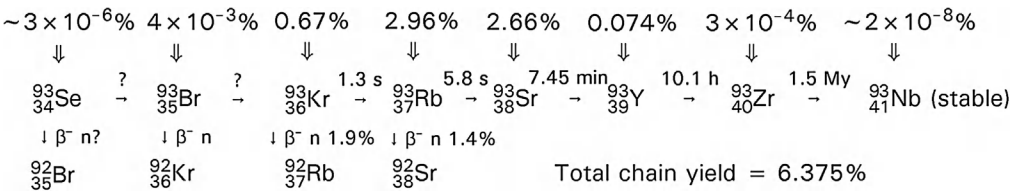


Figure 11.10 Fission product decay chain of $A = 93$. The independent yields in the upper row refer to nuclides believed to be formed directly in fission of ^{235}U by thermal neutrons.

units below the value of Z for stable nuclides with the same A. For $A = 93$ the yield is largest for $Z = 37$ and 38 (*most probable charge*, Z_p) compared to the stable value of $Z = 41$.

$$\gamma(A, Z) = \gamma(A)s^{-1}(2\pi)^{-1/2}e^{-(Z-Z_p)^2/(2s^2)} \tag{11.17}$$

where $\gamma(A,Z)$ is the initial yield of the fission fragment with mass A and atomic number Z, $\gamma(A)$ the total yield of mass A (usually given in % of all fissions) and s the width parameter for the charge distribution at mass A. Tables of $\gamma(A)$, Z_p and s for fission of several nuclides by thermal and energetic neutrons are available in the literature. Values of $\gamma(A)$, the *chain yield*, for fission of ^{235}U by thermal neutrons are usually also given in nuclide charts. However, it should be noted that several of the individual yields given in the data-bases or nuclide charts are not measured but calculated from a fitted model.

Fission of heavy elements other than uranium can be made to occur by particle bombardment, particularly if we use high energy neutrons or high energy charged particles such as protons. The mass distribution curve for this type of fission is interesting. At low bombarding energies it is asymmetric for many heavy elements as it is with low energy neutrons. However, as the energy of bombardment is increased the valley between the peaks of the curve becomes more shallow and, at high energies, a single-humped symmetric curve is obtained (Fig. 11.11). Thus, the most probable mode of mass split changes from asymmetric at low energies to symmetric at high energies. As the

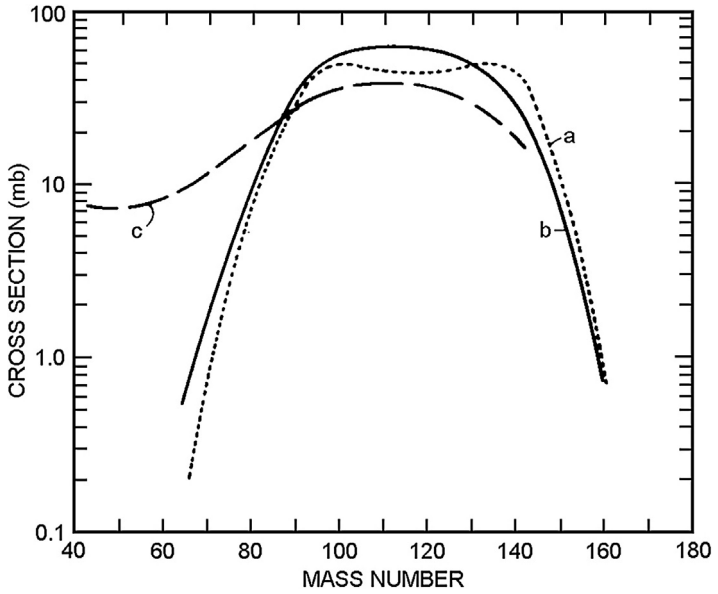


Figure 11.11 Mass yield curves for fission products from uranium irradiated with protons: (a) 100 MeV, (b) 170 MeV, and (c) 2.9 GeV. (From Friedlander.)

energy is increased, the fission yield curve $y(A)$ or cross-section curve $\sigma_{\text{fiss}}(A)$ becomes indistinguishable from yield curves of the type in Figure 11.6 ascribed to direct interaction mechanisms.

In contrast to the above, fission of nuclei in the radium region results in mass yield curves with three peaks symmetric around $\sim A/2$. Furthermore, low energy or spontaneous fission of some heavy actinide isotopes, e.g. ^{259}Fm, produces symmetric mass yield curves.

The light mass peak of double-humped fission yield curves shifts towards heavier masses when heavier nuclides undergo fission, but the position of the heavier mass peak remains almost constant. The Z_p-value increases somewhat with increasing charge of the fissioning nucleus. As an example the average mass of the light and heavy mass yield peaks in thermal fission of ^{235}U, ^{239}Pu and ^{241}Pu are 96.57, 100.34, 102.58 and 139.43, 139.66, 139.42 u, respectively, c.f. Fig. 11.9.

11.7.2. Energy of fission

From the curve of the binding energy per nucleon (see Fig. 4.3), we calculated in §4.4 that about 200 MeV would be released in the fission of a heavy element. In this section we consider how this fission energy is partitioned.

The neutrons emitted have an average kinetic energy of ~2 MeV. For the average of 2.5 neutrons emitted in fission of ^{235}U by thermal neutrons, about 5 MeV of the fission energy is required. The emission of γ-rays in the act of fission, *prompt* γ-rays, accounts for another 6—8 MeV. The largest part of the fission energy is observed as the kinetic energy of the fission products. We can estimate this by calculating the Coulombic repulsion energy of a probable fission product pair, $^{93}_{37}$Rb and $^{143}_{55}$Cs. The model used is two touching spherical nuclei with a charge center distance $d = 12.7$ fm (Fig. 11.12a). From (10.16) we calculate that the repulsive Coulomb energy (and, hence, the kinetic energy of separation) is 175 MeV. If the model is modified slightly to include a neck between the two nuclei, increasing the distance between the charge centers to 13.5 fm, the kinetic energy would be 165 MeV, as experiment requires. Such an elongated shape with a small neck at the time of separation in fission (the "scission" shape) is supported by several types of evidence.

The remaining ~23 MeV of fission energy is retained in the fission product nuclei as internal excitation and mass energy. This energy is released in a sequence of β-decay steps in which the N/Z values are adjusted to stability. Table 11.1 summarizes the distribution of fission energy.

11.7.3. Fragment kinetic energies

In fission induced by thermal neutrons (or spontaneous fission) the fissioning nucleus has a low kinetic energy. Thus the kinetic energies of the fragments are very near the same in the laboratory coordinate system and in the center-of-mass system. During fission, the

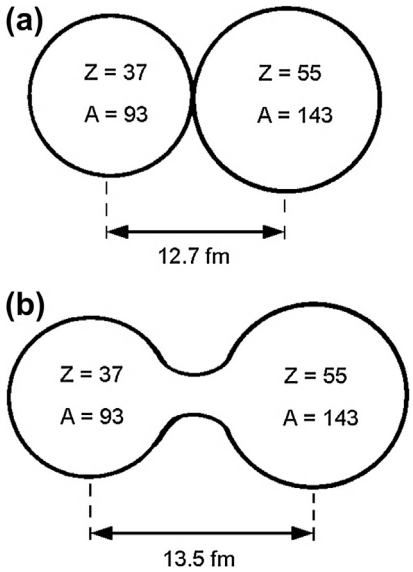


Figure 11.12 Fission of ^{236}U showing the distance between the effective centers of charge. (a) touching spheroids, (b) a scissioning dumbbell. Nuclear radii are calculated by $r = A^{1/3}$.

center of mass must remain stationary and the total impulse must be zero. Assuming $A \propto m$, the fragment kinetic energy, E_1, of a fragment with mass A_1 is given by

$$E_1 = [A_2/(A_1 + A_2)]E_{\text{kin}} \tag{11.18}$$

where E_{kin} is the total kinetic energy of the two fission fragments and $A_1 + A_2$ is the mass of the fissioning nucleus. Hence the lighter fragment initially carries a larger part of the kinetic energy than the heavier fragment. The initial distribution of kinetic energy is distorted by the emission of energetic neutrons shortly after fragment separation. Fission

Table 11.1 Data for energy distribution in thermal fission of ^{235}U in MeV

Prompt energy		*176.5 ± 5.5*
of which the	kinetic energy of fission products	164.6 ± 4.5
	kinetic energy of 2.5 prompt neutrons	4.9 ± 0.5
	γ-energy (prompt)	7.0 ± 0.5
Delayed energy from fission product decay		23.5 ± 5.0
of which the	kinetic energy of β's	6.5 ± 1.5
	neutrino radiation	10.5 ± 2.0
	γ-energy	6.5 ± 1.5
Total		200.0 ± 6

fragments are often used when calibrating the energy scale of heavy ion surface barrier detectors. The number-averaged fragment energies in spontaneous fission of ^{252}Cf are 105.71 MeV for light fragments and 80.01 MeV for heavy fragments. Rocket engines based on direct fission fragment emission have been comtemplated by NASA and seems today to be the only viable means to reach our nearest stars in less than a few 100 years.

11.7.4. Fission models

The analogy between nuclei and liquid droplets was found to be useful in deriving the semi-empirical mass formula (§4.6). Bohr and Wheeler explained fission just months after its discovery by using the same model. The surface tension of a liquid causes a droplet to assume a spherical shape, but if energy is supplied in some fashion this shape is distorted. If the attractive surface tension force is greater than the distorting force, the drop oscillates between spherical and elongated shapes. If, however, the distorting force becomes larger than the attractive force, the drop elongates past a threshold point and splits (fission).

In §4.6 we described how the repulsive forces between the protons in the nucleus could be expressed by a term a_c proportional to $Z^2/A^{1/3}$, and the surface tension attraction by another term a_s proportional to $A^{2/3}$. The repulsive Coulomb force tends to distort the nucleus in the same way a distorting force does a droplet, while the surface tension tries to bring it in to a spherical form. The ratio between the two opposing energies should measure the instability to fission of the nucleus. As shown in §6.7.4, the liquid drop model predicts that the probability of fission should increase with increasing Z^2/A. Of all naturally occurring nuclides only ^{235}U can be fissioned by thermal neutrons, while ^{238}U fission requires energetic neutrons ($\geq$ 2 MeV). With increasing Z ($>$ 92) the fission probability with thermal neutrons increases and the half-life of radioactive decay by spontaneous fission decreases. Both of these processes are more probable for even Z-elements than for odd Z-elements. The half-life for spontaneous fission decay is given in Figure 11.13 as a function of the fissionability parameter, x.

In the semi-empirical mass equation a spherical shape is assumed. If N and Z are kept constant and the potential energy of the nuclear liquid drop is calculated as a function of deformation from spherical to prolate, the curve in Figure 11.14a is obtained. The nucleus exists normally in the ground state level of the potential well. In order to undergo fission it must be excited above the fission barrier which is about 5–6 MeV. As the diagram shows, this means excitation of the nucleus into the continuum level region if the nucleus retains the shape associated with the potential well. However, if the nucleus deforms, some excitation energy goes into deformation energy. At the top of the barrier, the nucleus is highly deformed and has relatively little internal excitation energy. It exists in well-defined vibrational levels, and fission occurs from such a level. This is known as the "saddle point" (the top of the barrier) of fission.

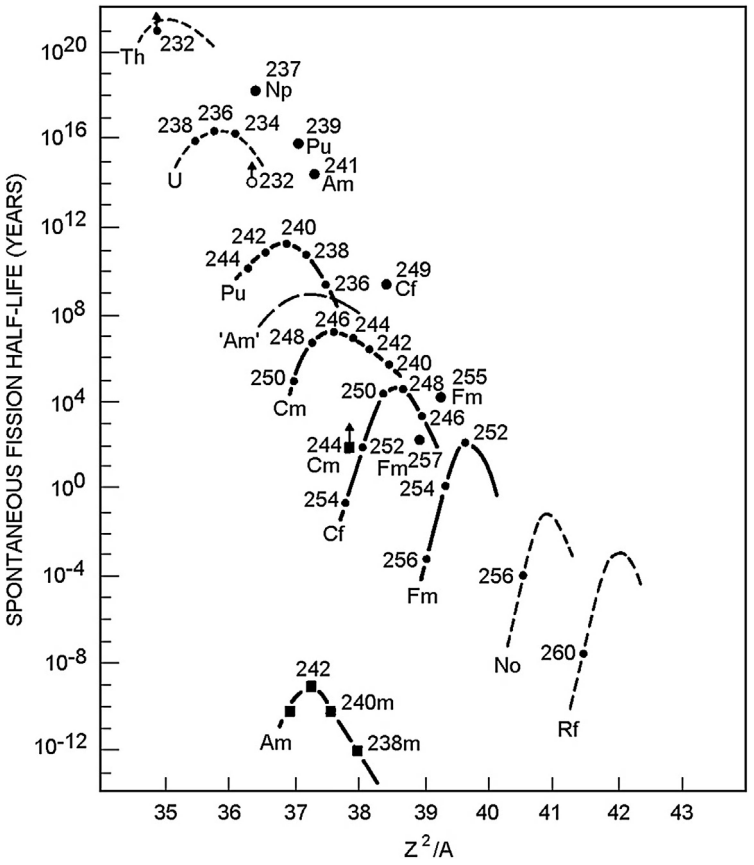


Figure 11.13 Spontaneous fission lifetimes. (From Strutinsky and Bjørnholm.)

It has long been recognized that the liquid-drop model semi-empirical mass equation cannot calculate the correct masses in the vicinity of neutron and proton magic numbers. More recently it was realized that it is less successful also for very deformed nuclei midway between closed nucleon shells. Introduction of magic numbers and deformations in the liquid drop model improved its predictions for deformed nuclei and of fission barrier heights. However, an additional complication with the liquid-drop model arose when isomers were discovered which decayed by spontaneous fission. Between uranium and californium a number of nuclides were found to decay by spontaneous fission with half-lives of 10^{-2} to 10^{-9} s, millions of times slower than prompt fission which occurs within 10^{-14} s but millions of times faster than normal spontaneous fission (Fig. 11.13). For example, ^{242}Cm has a ground state half-life to spontaneous fission of 10^6 years, while an isomeric state of ^{242}Cm has been found to fission with $t_{½}$ of 10^{-7} s.

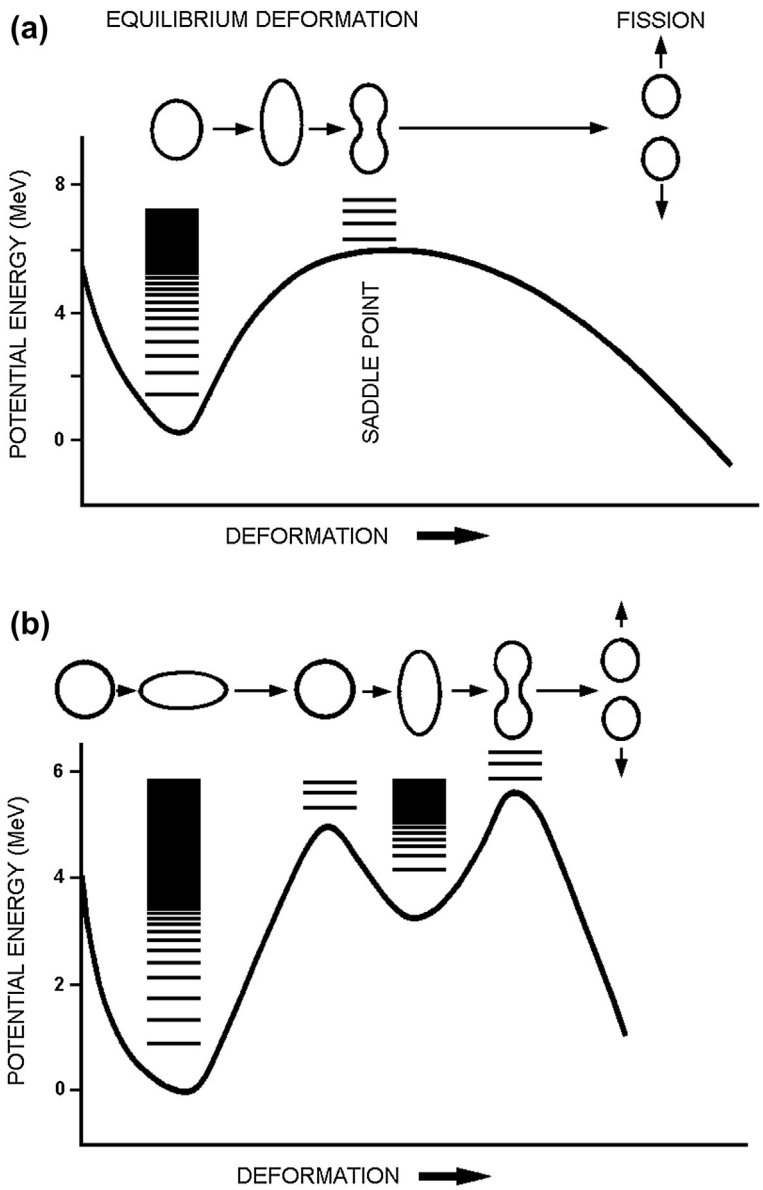


Figure 11.14 (a) The liquid-drop model potential energy curve. (b) Same, but modified by shell corrections.

Strutinsky developed an extension of the liquid drop model which satisfactorily explains the fission isomers and asymmetric fission. For such short half-lives the barrier must be only 2−3 MeV. Noting the manner in which the shell model levels vary with deformation (§6.5, the "Nilsson levels"), Strutinsky added shell corrections to the basic

liquid-drop model and obtained the "double-well" potential energy curve in Figure 11.14b. In the first well the nucleus is a spheroid with the major axis about 25% larger than the minor. In the second well, the deformation is much larger, the axis ratio being about 1.8. A nucleus in the second well is metastable (i.e. in isomeric state) as it is unstable to γ-decay to the first well or to fission. Fission from the second well is hindered by a 2−3 MeV barrier, while from the first well the barrier is 5−6 MeV, accounting for the difference in half-lives.

The single-well curve in Figure 11.14a predicts symmetric fission whereas the double-well curve (Fig. 11.14b) leads to the correct prediction of asymmetric fission and a thin neck. Incorporation of shell effects in the fission model also leads to the prediction that the half-lives of very heavy nuclides ($Z \geq 106$) must be longer than the simple liquid-drop model would indicate. This led to a search for "super heavy" elements with $Z = 110 - 118$. The major difficulty is the release of reaction energy which causes immediate fission of the compound nucleus. Hence, strongly endothermic reactions have been tried. However, these have extremely small reaction yields. The heaviest man made elements are discussed in Ch. 14.

11.8. PHOTONUCLEAR REACTIONS

If a photon transfers sufficient energy to a nucleus to excite it to a higher state, three possibilities for de-excitation exist: (a) the same energy is immediately re-emitted isotropically, (b) a long-lived isomer may be formed which decays through emission of one or more γ-rays, and (c) the nucleus disintegrates. The first process (a) is referred to as the Mössbauer effect and is discussed in chapter 7. The second process (b) of nuclear de-excitation has also been discussed earlier. The third process (c) is referred to as *photonuclear disintegration*. The energy transferred to the nucleus must be sufficient to excite it above the dissociation energy for a proton, neutron, or other particle. A large energy transfer can also induce fission of heavy nuclei, *photo fission*.

The simplest photodisintegration process is that of the deuteron, whose binding energy is 2.23 MeV. If the γ-ray energy absorbed exceeds this value, a neutron and a proton are formed. This is a common reaction in nuclear reactors using heavy water as moderator, because the fission γ-ray energy is often several MeV. However, γ-rays of such energy rarely occur in the radioactive decay of nuclides (i.e. with half-lives of hours or longer). The cross-section for photodisintegration of ^{2}H has a maximum value of 2.4 mb at 4.3 MeV E_γ.

The energy necessary for photodisintegration of a nucleus is calculated from known nuclear masses. It is obviously easier to remove one particle than several from a nucleus. As a result we find that E_γ must be ≥ 5 MeV for photodisintegration of heavier nuclei.